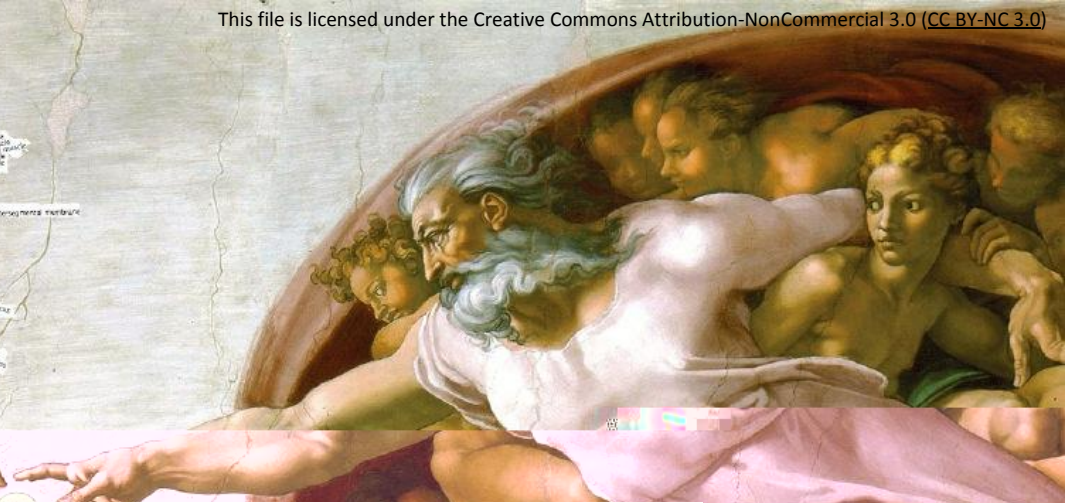




Lecture 1 - Knowledge Graphs in the Web of Data

1.3 The Art of Understanding

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Knowledge Graphs

Lecture 1: Knowledge Graphs in the Web of Data

1.1 Data, Information, and Knowledge

1.2 How to Represent Knowledge?

1.3 The Art of Understanding

1.4 Towards a Universal Knowledge Representation

1.5 The Semantic Web

1.6 Linked Data and the Web of Data

Climate Change is
the Everest of all



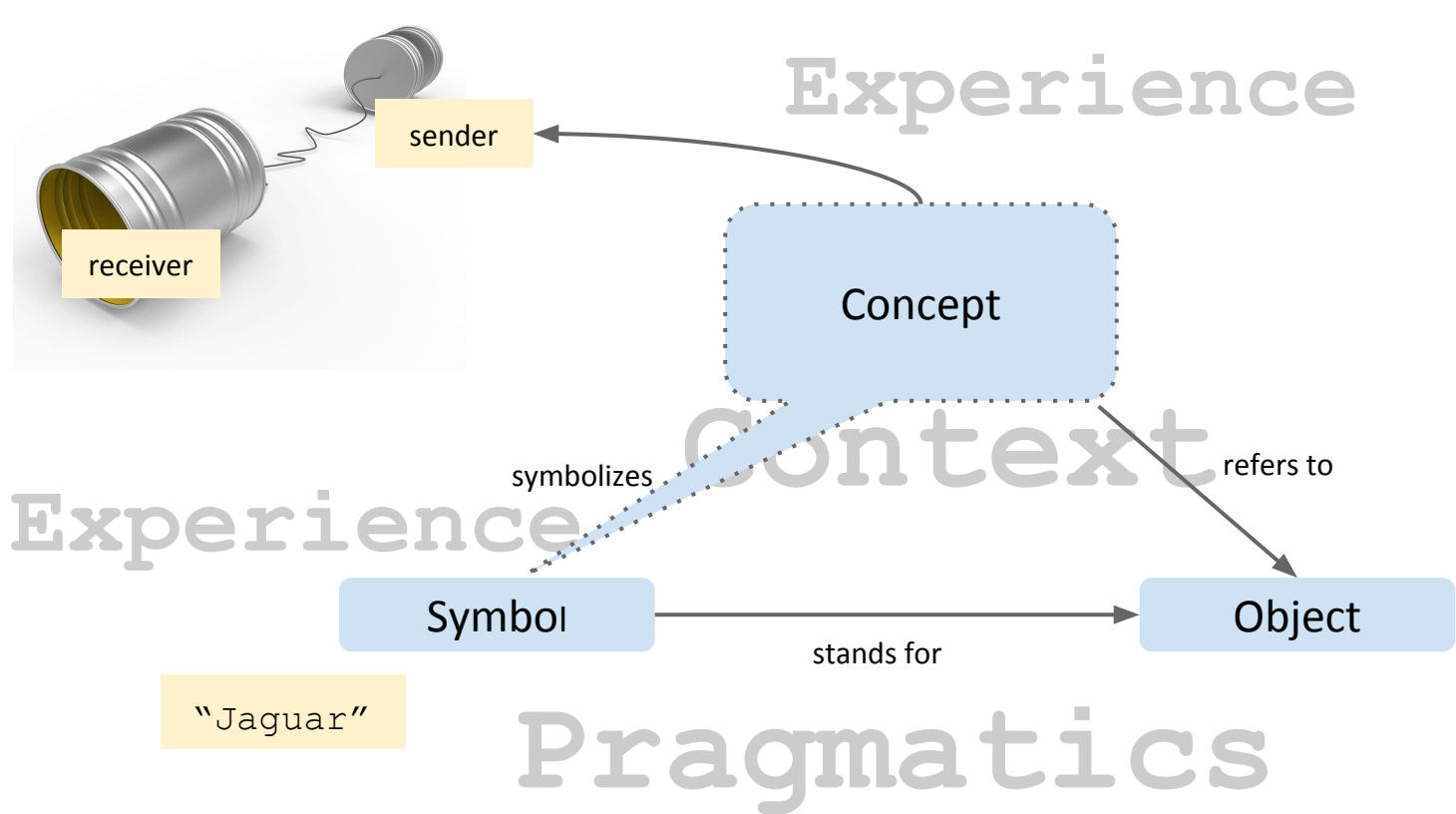
What does it mean to “understand”?

- Understanding (in general) is the ability to grasp the meaning of information.
- Information is conveyed in a message using a specific language from a sender to a receiver.



- Information is understood by the receiver of a message, if the receiver interprets the information correctly.

Meaning and Understanding



[2,3,4,5]



Meaning and Understanding

- Correct Interpretation depends on
 - Syntax,
 - Semantics,
 - Context,
 - Pragmatics, and
 - Experience.

Syntax

- =[greek] Arrangement, Ordering
- In grammatics syntax denotes the study of the principles and processes by which sentences are constructed in particular languages.
- In formal languages, syntax is just a set of rules, by which well formed expressions can be created from a fundamental set of symbols (alphabet).
- In computer science, syntax defines the normative structure of data.

Semantics

- =[greek] pertains to the character, the study of meaning
- is part of the linguistics which focuses on Sense and Meaning of language or symbols of language.
- is the study of interpretation of signs or symbols as used by agents or communities within particular circumstances and contexts.
- Semantics asks, how sense and meaning of complex concepts can be derived from simple concepts based on the rules of syntax.
- The semantics of a message depends on context and pragmatics.

Context

- [lat.] contextus = interweaved
- denotes the surrounding of a symbol (concept) in an expression resp. its relationship with surrounding expressions (concepts) and further related elements,
- Context denotes all elements of any sort of communication that define the interpretation of the communicated content.

Pragmatics

- =[greek] action
- reflects the intention by which the language is used to communicate a message.
- In linguistics, pragmatics denotes the study of applying language in different situations.
- It also denotes the intended purpose of the speaker.
- Pragmatics studies the ways in which context contributes to meaning.

Experience

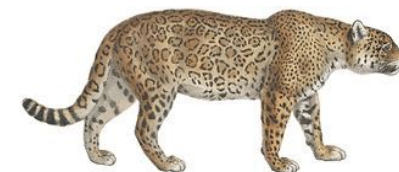
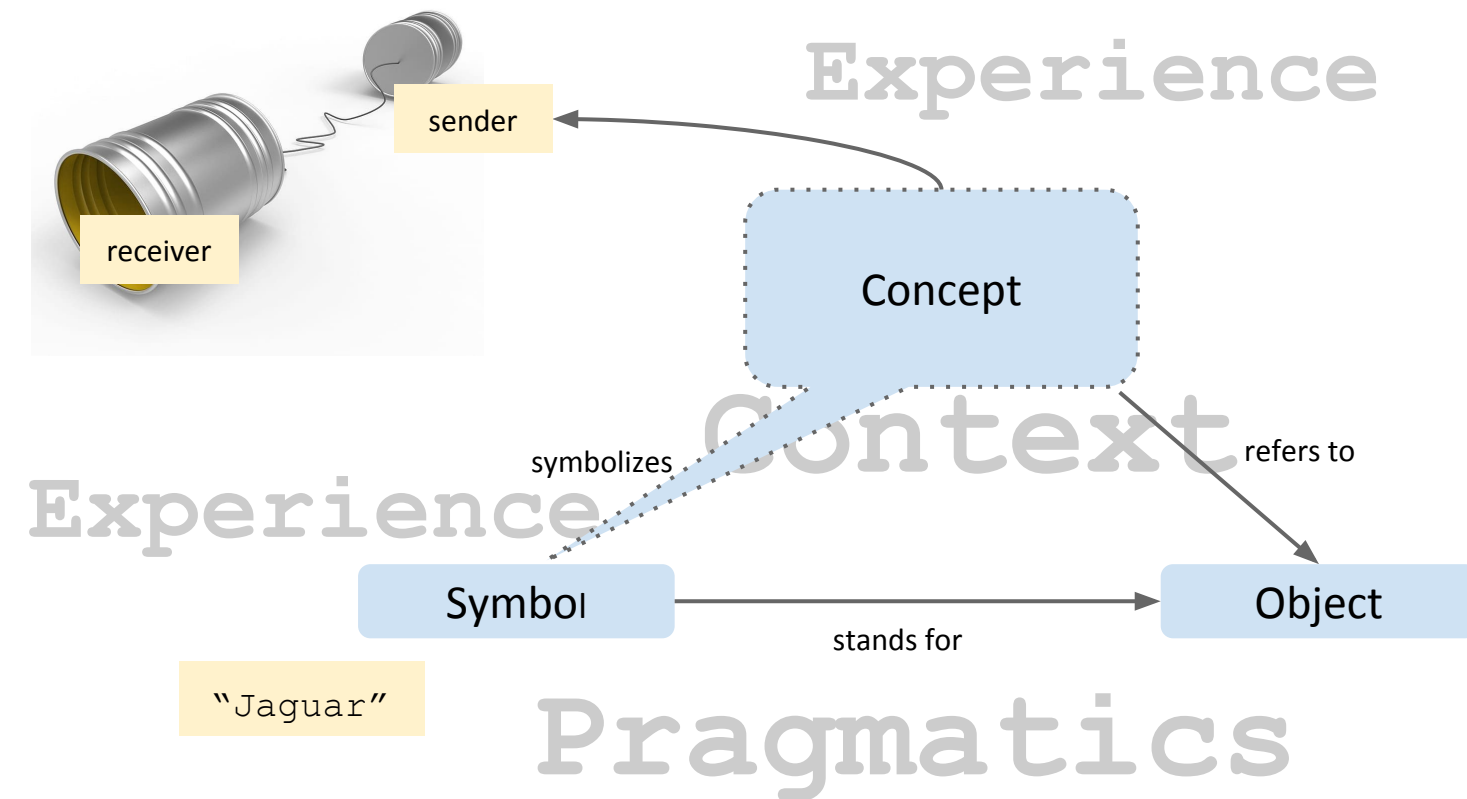
- Experience considers all information that you have learned and put in context with the world you are living in.
- Experience in this sense is often referred to as common sense knowledge or world knowledge.

Successful Communication

- For successful communication,
 - information has to be correctly transmitted (Syntax)
 - the meaning (Semantics) of the transmitted information must be interpreted correctly (= understanding)
- Understanding depends on
 - the context of both sender and receiver and
 - the pragmatics of the sender
- (Personal) experience determines
 - how sender and receiver interpret the semantics, context, and pragmatics of a message, and thus its intended meaning

[2,3,4,5]

Meaning and Understanding



Towards a Universal Knowledge Representation

Next Lecture...

Picture References:

- [1] Mount Everest as seen from an aircraft from airline company Drukair in Bhutan.
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